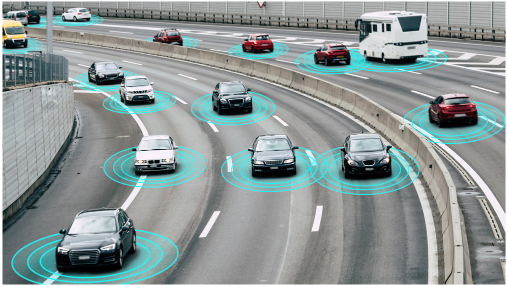
Quantifying the Reproducibility of Multi-Band High Speed Wireless Channel Measurements

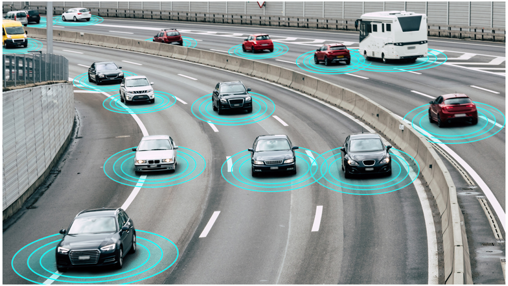
F. Pasic, M. Hofer, D. Radovic, H. Groll, S. Caban,
T. Zemen and C. F. Mecklenbräuker

Institute for Telecommunications, TU Wien

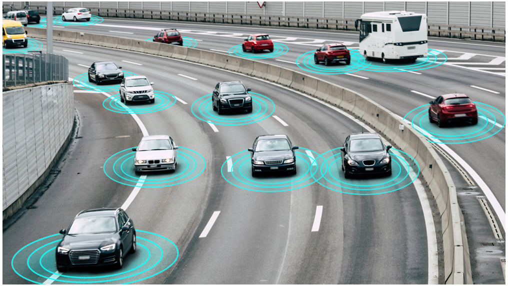
- increasing demand for high data rate transmission



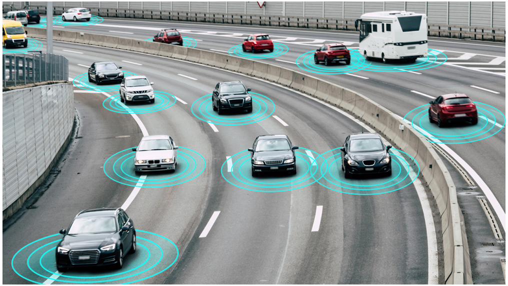
- increasing demand for high data rate transmission
  - high-mobility environments



- increasing demand for high data rate transmission
  - high-mobility environments
  - sub-6 GHz band cannot meet these requirements

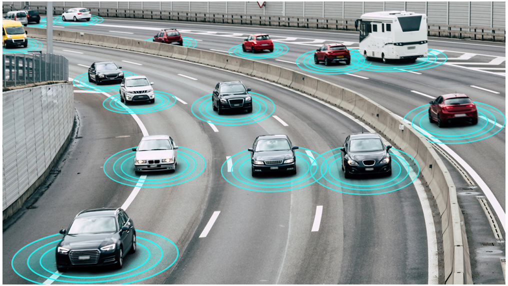


- increasing demand for high data rate transmission
  - high-mobility environments
  - sub-6 GHz band cannot meet these requirements
- mmWave bands allow for significantly higher data rates



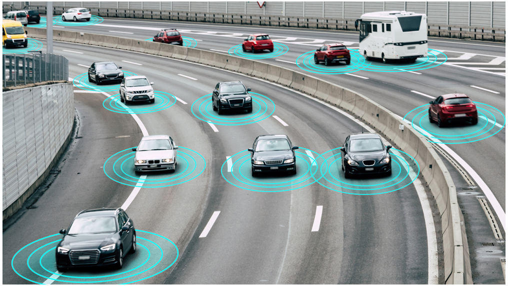
Introduction

- increasing demand for high data rate transmission
  - high-mobility environments
  - sub-6 GHz band cannot meet these requirements

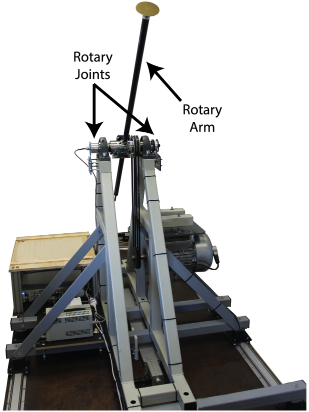


- mmWave bands allow for significantly higher data rates
- understand differences and similarities in terms of propagation channel characteristics

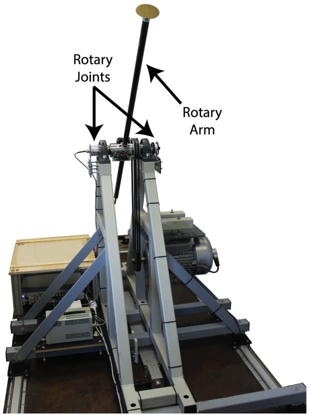
- increasing demand for high data rate transmission
  - high-mobility environments
  - sub-6 GHz band cannot meet these requirements
- mmWave bands allow for significantly higher data rates
- understand differences and similarities in terms of propagation channel characteristics
- perform measurements and compare these bands in a fair manner



- measurement setup and methodology
  - reproducible measurements
  - identical Tx and Rx antenna positions

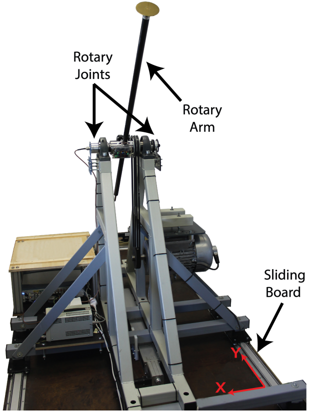


- measurement setup and methodology
  - reproducible measurements
  - identical Tx and Rx antenna positions
- moving transmitter and statical receiver
  - rotary unit
  - trigger unit



- measurement setup and methodology
  - reproducible measurements
  - identical Tx and Rx antenna positions

- moving transmitter and statical receiver
  - rotary unit
  - trigger unit

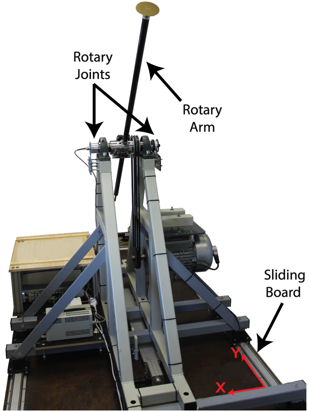


Measurement Setup

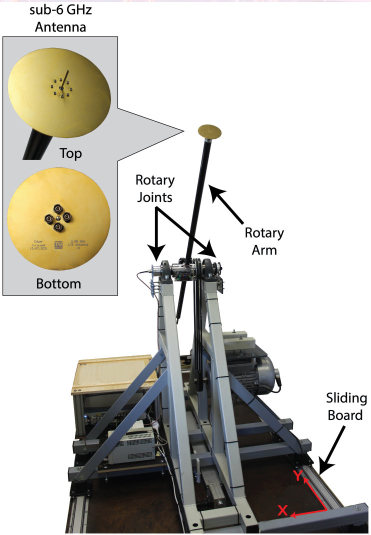
- measurement setup and methodology
  - reproducible measurements
  - identical Tx and Rx antenna positions

- moving transmitter and statical receiver
  - rotary unit
  - trigger unit

- operates at 2.55 GHz and 25.5 GHz

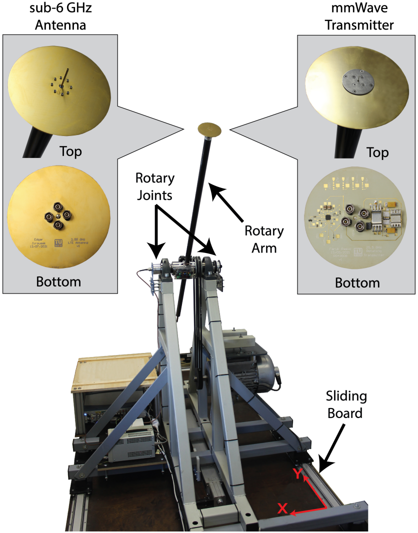


- measurement setup and methodology
  - reproducible measurements
  - identical Tx and Rx antenna positions
- moving transmitter and statical receiver
  - rotary unit
  - trigger unit
- operates at 2.55 GHz and 25.5 GHz
  - sub-6 GHz: direct transmission



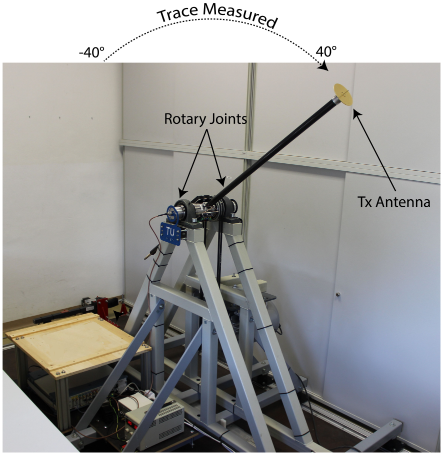
Measurement Setup

- measurement setup and methodology
  - reproducible measurements
  - identical Tx and Rx antenna positions
- moving transmitter and statical receiver
  - rotary unit
  - trigger unit
- operates at 2.55 GHz and 25.5 GHz
  - sub-6 GHz: direct transmission
  - mmWave: lightweight transmitter



Measurement Setup

- measurement setup and methodology
  - reproducible measurements
  - identical Tx and Rx antenna positions
- moving transmitter and statical receiver
  - rotary unit
  - trigger unit
- operates at 2.55 GHz and 25.5 GHz
  - sub-6 GHz: direct transmission
  - mmWave: lightweight transmitter
- perform wireless channel measurements along the same trace for different frequency bands



- can experiments be faithfully reproduced ?

Motivation

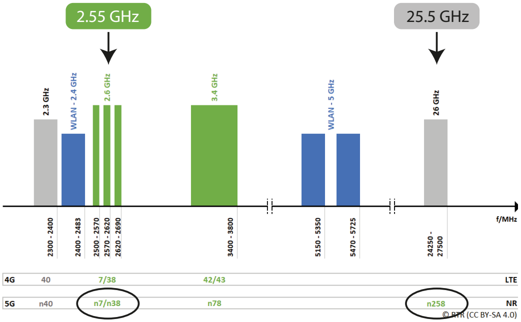
2.55 GHz

25.5 GHz

- can experiments be faithfully reproduced ?
- undesired factors
  - interference from n7/n38 and n258

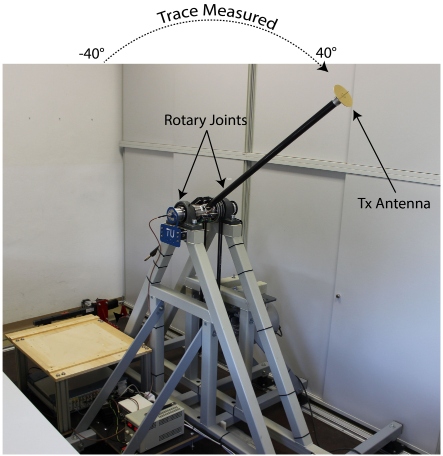
Motivation

- can experiments be faithfully reproduced ?
- undesired factors
  - interference from n7/n38 and n258

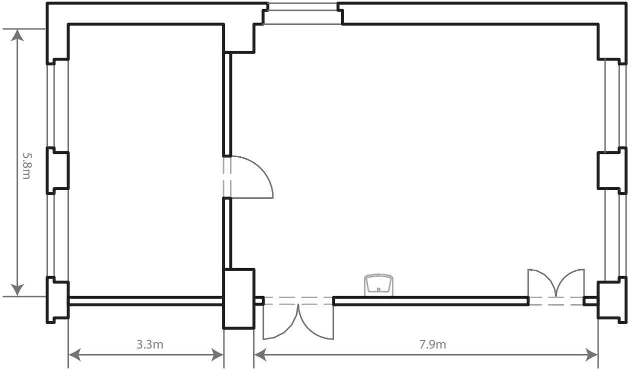


Motivation

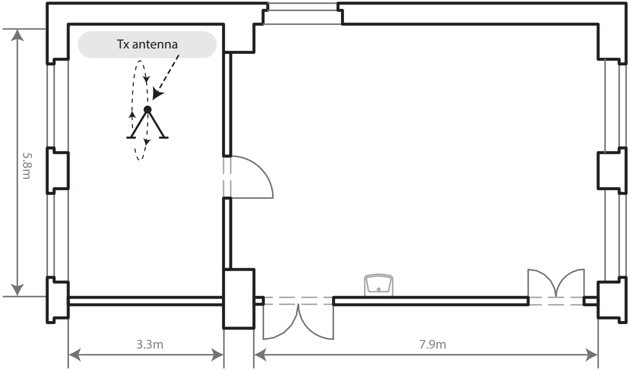
- can experiments be faithfully reproduced ?
- undesired factors
  - interference from n7/n38 and n258
  - equipment malfunction



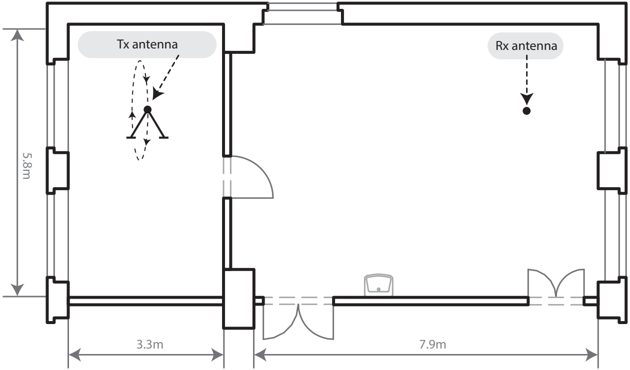
- wireless channel measurements



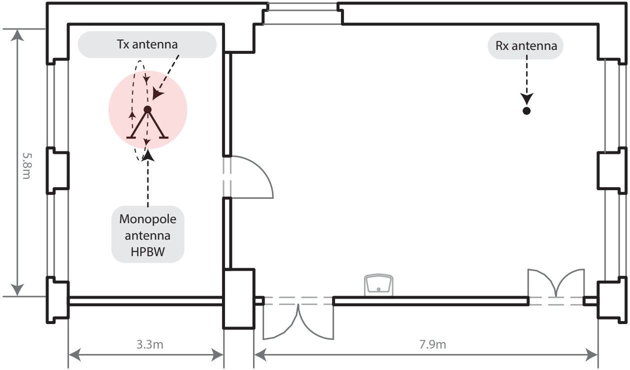
- wireless channel measurements
  - moving Tx antenna (50 km/h)



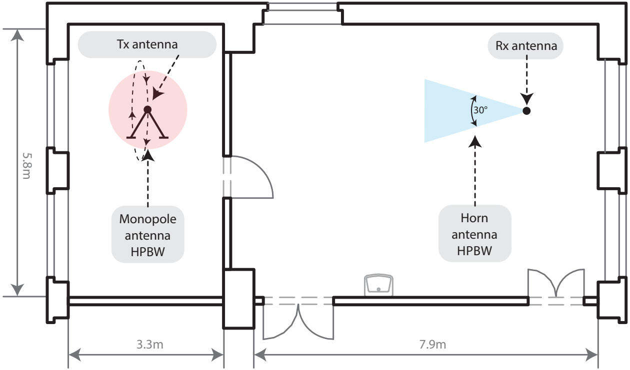
- wireless channel measurements
  - moving Tx antenna (50 km/h)
  - static Rx antenna



- wireless channel measurements
  - moving Tx antenna (50 km/h)
  - static Rx antenna
  - Tx: monopole antenna



- wireless channel measurements
  - moving Tx antenna (50 km/h)
  - static Rx antenna
  - Tx: monopole antenna
  - Rx: horn antenna



Measurement Campaign

- wireless channel measurements
  - moving Tx antenna (50 km/h)
  - static Rx antenna
  - Tx: monopole antenna
  - Rx: horn antenna
- measurement procedure
  - channel sounding using OFDM

Parameter	Value
Total Bandwidth	$B_{\mathrm{t}} = 100\,\mathrm{MHz}$
Number of Subcarriers	$K = 200$
number of repetitions	$T = 100$
number of snapshots	$N = 500$
symbols per snapshot	$N_{\mathrm{sym}} = 100$
subcarrier spacing	$\triangle f = 500\,\mathrm{kHz}$
transmitter velocity	$v_{\mathrm{Tx}} = 50\,\mathrm{km/h}$
symbol duration	$t_{\mathrm{s}} = 2\,\mu\mathrm{s}$
snapshot duration	$t_{\mathrm{snap}} = 200\,\mu\mathrm{s}$
transmit power	$P_{\mathrm{Tx}} = 10\,\mathrm{dBm}$
transmit antenna	5 dBi vertical monopole antenna

Measurement Campaign

- wireless channel measurements
  - moving Tx antenna (50 km/h)
  - static Rx antenna
  - Tx: monopole antenna
  - Rx: horn antenna
- measurement procedure
  - channel sounding using OFDM
  - 100 identical experiments
  - center frequency: 2.55 GHz or 25.5 GHz

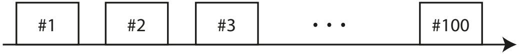
Parameter	Value
Total Bandwidth	$B_{\mathrm{t}} = 100\,\mathrm{MHz}$
Number of Subcarriers	$K = 200$
number of repetitions	$T = 100$
number of snapshots	$N = 500$
symbols per snapshot	$N_{\mathrm{sym}} = 100$
subcarrier spacing	$\triangle f = 500\,\mathrm{kHz}$
transmitter velocity	$v_{\mathrm{Tx}} = 50\,\mathrm{km/h}$
symbol duration	$t_{\mathrm{s}} = 2\,\mathrm{\mu s}$
snapshot duration	$t_{\mathrm{snap}} = 200\,\mathrm{\mu s}$
transmit power	$P_{\mathrm{Tx}} = 10\,\mathrm{dBm}$
transmit antenna	5 dBi vertical monopole antenna

- wireless channel measurements
  - moving Tx antenna (50 km/h)
  - static Rx antenna
  - Tx: monopole antenna
  - Rx: horn antenna
- measurement procedure
  - channel sounding using OFDM
  - 100 identical experiments
  - center frequency: 2.55 GHz or 25.5 GHz
- result: time-variant channel transfer function

Parameter	Value
Total Bandwidth	$B_{\mathrm{t}} = 100\,\mathrm{MHz}$
Number of Subcarriers	$K = 200$
number of repetitions	$T = 100$
number of snapshots	$N = 500$
symbols per snapshot	$N_{\mathrm{sym}} = 100$
subcarrier spacing	$\triangle f = 500\,\mathrm{kHz}$
transmitter velocity	$v_{\mathrm{Tx}} = 50\,\mathrm{km/h}$
symbol duration	$t_{\mathrm{s}} = 2\,\mathrm{\mu s}$
snapshot duration	$t_{\mathrm{snap}} = 200\,\mathrm{\mu s}$
transmit power	$P_{\mathrm{Tx}} = 10\,\mathrm{dBm}$
transmit antenna	5 dBi vertical monopole antenna

Contents

- time
  - similarity between transmissions



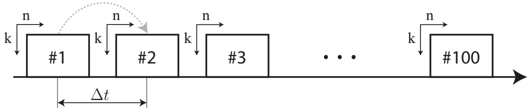
n - snapshots
k - subcarriers

- time
  - similarity between transmissions

- time
  - similarity between transmissions

$$r_{\Delta t} = \frac{\left|\mathrm{tr}\left(\mathbf{H}_t^{\mathrm{H}}\mathbf{H}_{t+\Delta t}\right)\right|}{\|\mathbf{H}_t\|_F\|\mathbf{H}_{t+\Delta t}\|_F}$$

n - snapshots
k - subcarriers

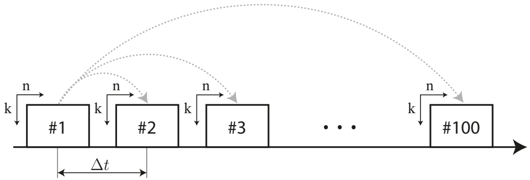


Results and Discussion

- time
  - similarity between transmissions

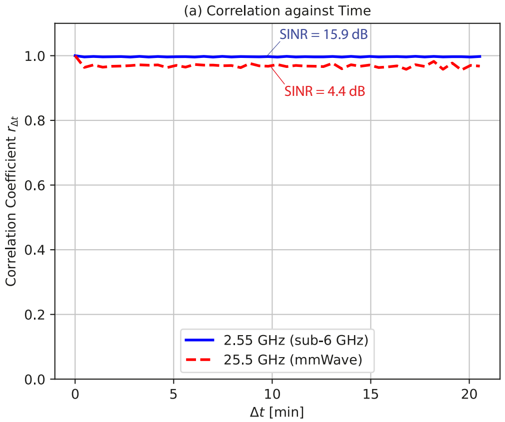
$$r_{\Delta t} = \frac{\left|\mathrm{tr}\left(\mathbf{H}_t^{\mathrm{H}}\mathbf{H}_{t+\Delta t}\right)\right|}{\|\mathbf{H}_t\|_F \|\mathbf{H}_{t+\Delta t}\|_F}$$

n - snapshots
k - subcarriers



Results and Discussion

- time
  - similarity between transmissions
  - system can be considered as stable



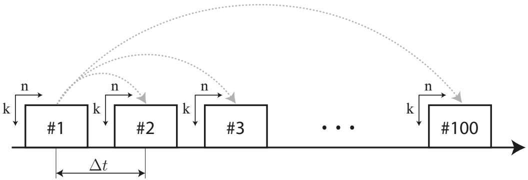
Results and Discussion

- time
  - similarity between transmissions
  - system can be considered as stable

- SINR
  - add complex Gaussian noise

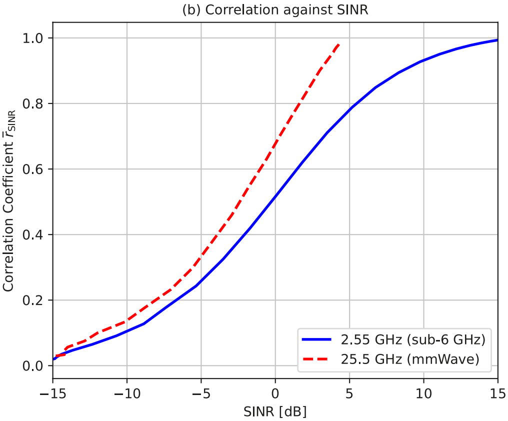
$$\overline{r}_{\text{SINR}} = \frac{1}{T} \sum_{\Delta t=0}^{T-1} \frac{\left| \text{tr}\left(\mathbf{H}_{t,\text{SINR}}^{\text{H}} \mathbf{H}_{t+\Delta t,\text{SINR}} \right) \right|}{\left\| \mathbf{H}_{t,\text{SINR}} \right\|_F \left\| \mathbf{H}_{t+\Delta t,\text{SINR}} \right\|_F}$$

n - snapshots
k - subcarriers



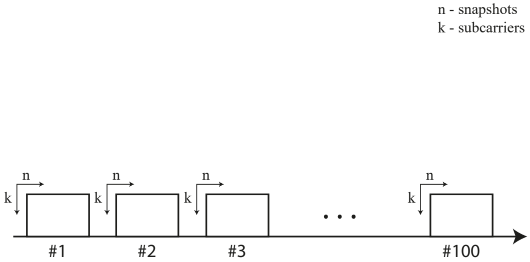
Results and Discussion

- time
  - similarity between transmissions
  - system can be considered as stable
- SINR
  - add complex Gaussian noise
  - correlation decreases with SINR
  - mmWave exhibit larger correlation

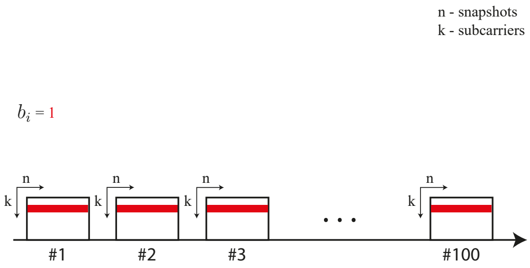


Results and Discussion

- time
  - similarity between transmissions
  - system can be considered as stable
- SINR
  - add complex Gaussian noise
  - correlation decreases with SINR
  - mmWave exhibit larger correlation
- bandwidth
  - calculation procedure



- time
  - similarity between transmissions
  - system can be considered as stable
- SINR
  - add complex Gaussian noise
  - correlation decreases with SINR
  - mmWave exhibit larger correlation
- bandwidth
  - calculation procedure

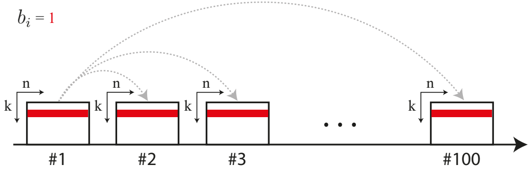


Results and Discussion

- time
  - similarity between transmissions
  - system can be considered as stable
- SINR
  - add complex Gaussian noise
  - correlation decreases with SINR
  - mmWave exhibit larger correlation
- bandwidth
  - calculation procedure

$$\bar{r}_B = \frac{1}{RT}\sum_{b_i=1}^{R}\sum_{\Delta t=0}^{T-1}\frac{\left|\operatorname{tr}\left(\mathbf{H}_{t,b_i}^{\mathrm{H}}\mathbf{H}_{t+\Delta t,b_i}\right)\right|}{\|\mathbf{H}_{t,b_i}\|_F\|\mathbf{H}_{t+\Delta t,b_i}\|_F}$$

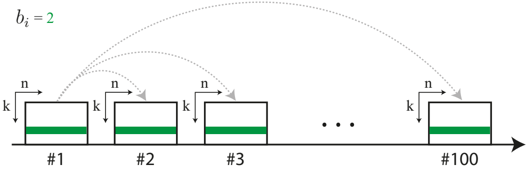
n - snapshots
k - subcarriers



- time
  - similarity between transmissions
  - system can be considered as stable
- SINR
  - add complex Gaussian noise
  - correlation decreases with SINR
  - mmWave exhibit larger correlation
- bandwidth
  - calculation procedure

$$\bar{r}_B = \frac{1}{RT} \sum_{b_i=1}^{R} \sum_{\Delta t=0}^{T-1} \frac{\left|\mathrm{tr}\left(\mathbf{H}_{t,b_i}^{\mathrm{H}} \mathbf{H}_{t+\Delta t,b_i}\right)\right|}{\|\mathbf{H}_{t,b_i}\|_F \|\mathbf{H}_{t+\Delta t,b_i}\|_F}$$

n - snapshots
k - subcarriers

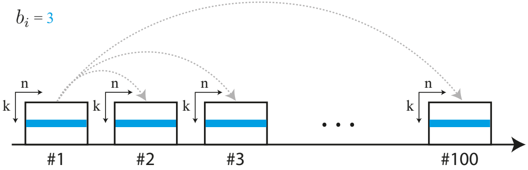


Results and Discussion

- time
  - similarity between transmissions
  - system can be considered as stable
- SINR
  - add complex Gaussian noise
  - correlation decreases with SINR
  - mmWave exhibit larger correlation
- bandwidth
  - calculation procedure

$$\bar{r}_B = \frac{1}{RT} \sum_{b_i=1}^{R} \sum_{\Delta t=0}^{T-1} \frac{\left|\operatorname{tr}\left(\mathbf{H}_{t,b_i}^{\mathrm{H}} \mathbf{H}_{t+\Delta t,b_i}\right)\right|}{\|\mathbf{H}_{t,b_i}\|_F \|\mathbf{H}_{t+\Delta t,b_i}\|_F}$$

n - snapshots
k - subcarriers

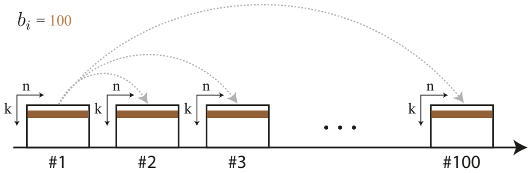


Results and Discussion

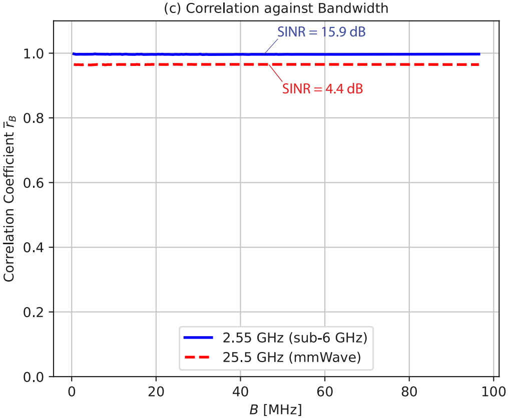
- time
  - similarity between transmissions
  - system can be considered as stable
- SINR
  - add complex Gaussian noise
  - correlation decreases with SINR
  - mmWave exhibit larger correlation
- bandwidth
  - calculation procedure

$$\bar{r}_B = \frac{1}{RT}\sum_{b_i=1}^{R}\sum_{\Delta t=0}^{T-1}\frac{\left|\mathrm{tr}\left(\mathbf{H}_{t,b_i}^{\mathrm{H}}\mathbf{H}_{t+\Delta t,b_i}\right)\right|}{\|\mathbf{H}_{t,b_i}\|_F\|\mathbf{H}_{t+\Delta t,b_i}\|_F}$$

n - snapshots
k - subcarriers



- time
  - similarity between transmissions
  - system can be considered as stable
- SINR
  - add complex Gaussian noise
  - correlation decreases with SINR
  - mmWave exhibit larger correlation
- bandwidth
  - calculation procedure
  - marginal variations over bandwidth



Summary and Conclusion

- reproducibility depends strongly on the measured SINR

- reproducibility depends strongly on the measured SINR
- reproducibility does not depend on the experiment duration and the system bandwidth

- reproducibility depends strongly on the measured SINR
- reproducibility does not depend on the experiment duration and the system bandwidth
- **proposed testbed is able to reproduce an experiment with minor uncertainties**

Thank you!

faruk.pasic@tuwien.ac.at